Hellenic Complex Systems Laboratory

Receiver Operating Characteristic Curves and Uncertainty of Measurement

Technical Report V

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2007

Receiver Operating Characteristic Curves and Uncertainty of Measurement

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Short Description of the Demonstration

This Demonstration compares two receiver operating characteristic (ROC) curves for two diagnostic tests measuring the same measurand (first test: blue plot, second test: orange plot). The comparisons are made for normally distributed nondiseased and diseased populations, for various values of the population means and standard deviations, and for different values of the tests' measurement uncertainty. A normal distribution of measurement uncertainty is assumed. The ratio of the areas under the two ROC curves is calculated. The six parameters that the user can vary using the sliders are measured in arbitrary units.

Search Terms

receiver operating characteristic curves, ROC curves, uncertainty of measurement, sensitivity, specificity, diagnostic test, diagnostic accuracy, permissible uncertainty, normal distribution, area under ROC curves

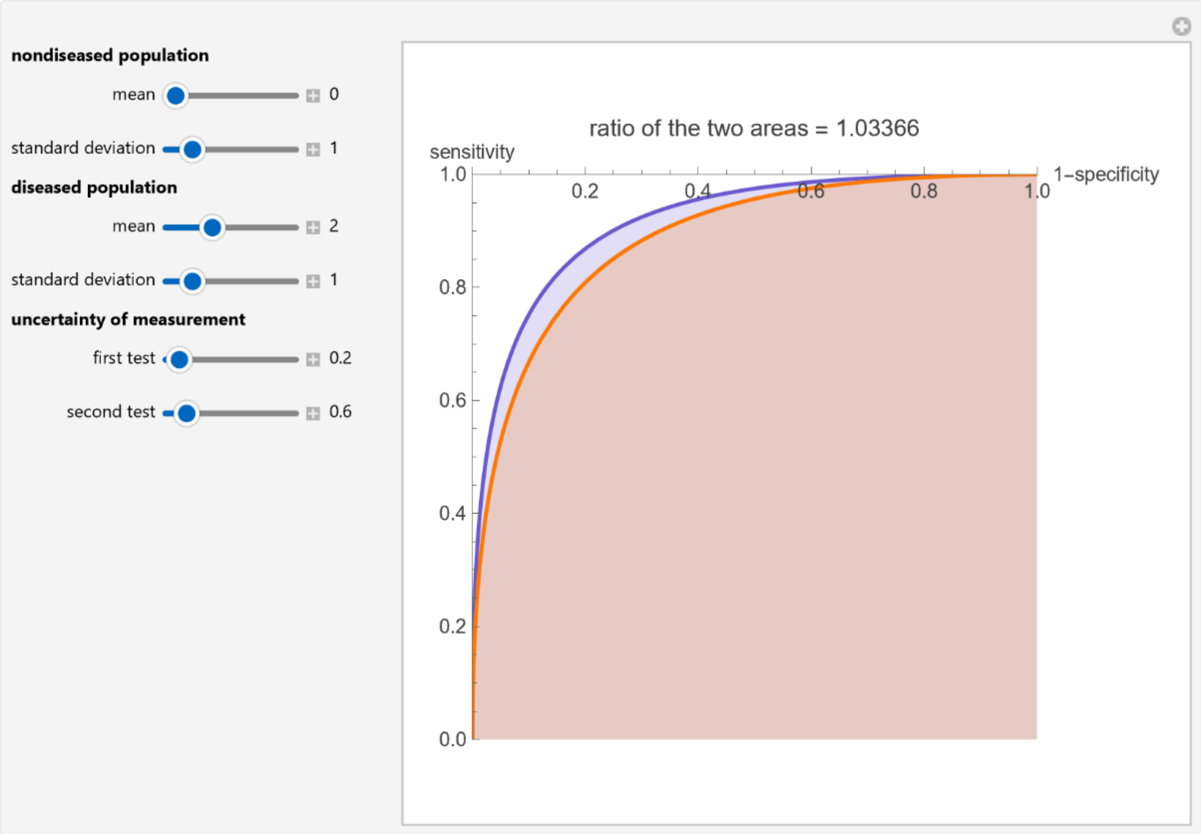

Figure 1: *Receiver operating characteristic (ROC) plots of two diagnostic tests (first test: blue plot, second test: orange plot), and the ratio of the respective areas under the curves, with the settings shown at the left.*

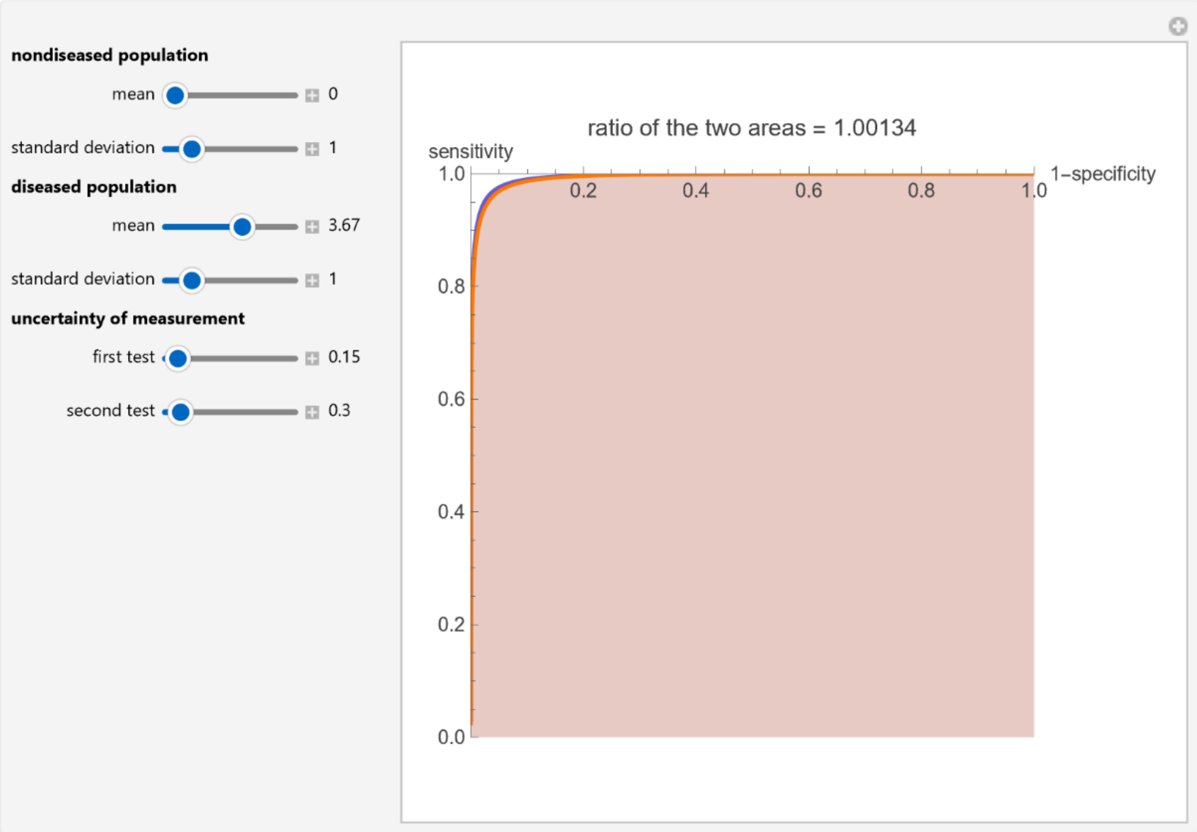

Figure 2: *Receiver operating characteristic (ROC) plots of two diagnostic tests (first test: blue plot, second test: orange plot), and the ratio of the respective areas under the curves, with the settings shown at the left.*

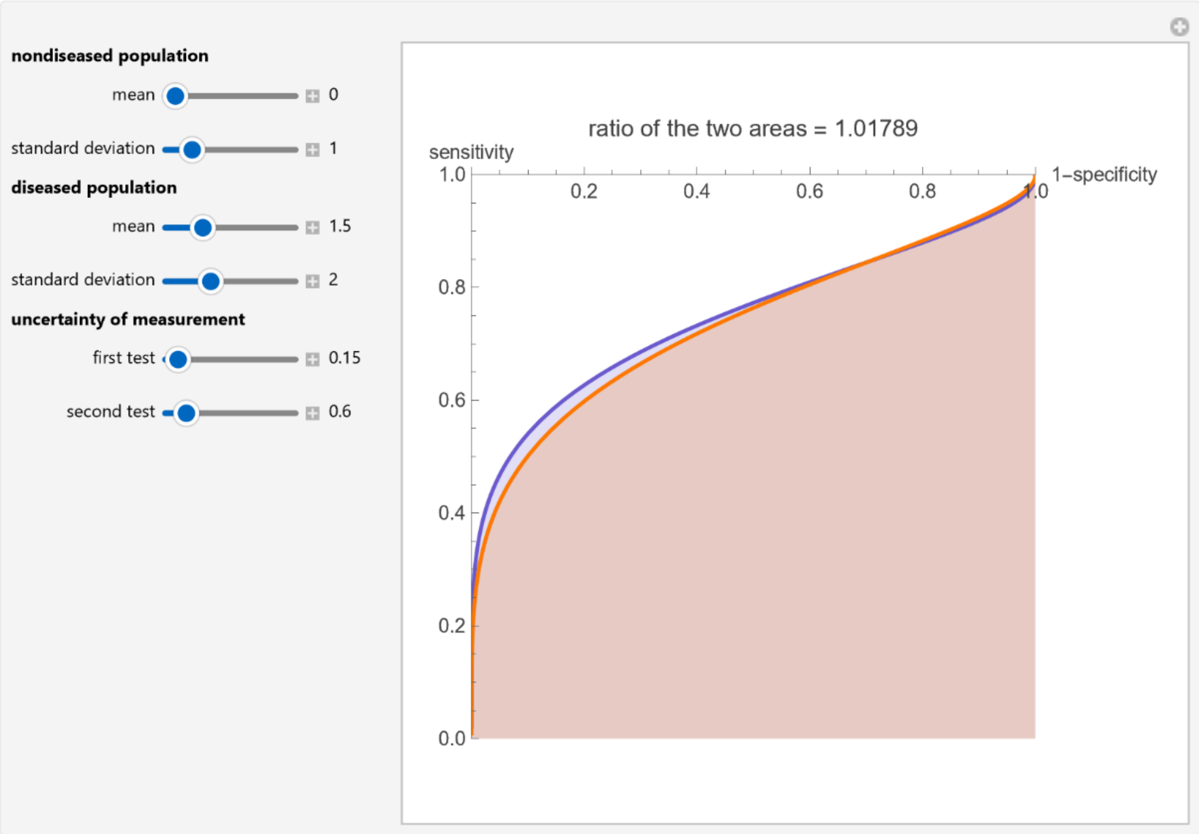

Figure 3: *Receiver operating characteristic (ROC) plots of two diagnostic tests (first test: blue plot, second test: orange plot), and the ratio of the respective areas under the curves, with the settings shown at the left.*

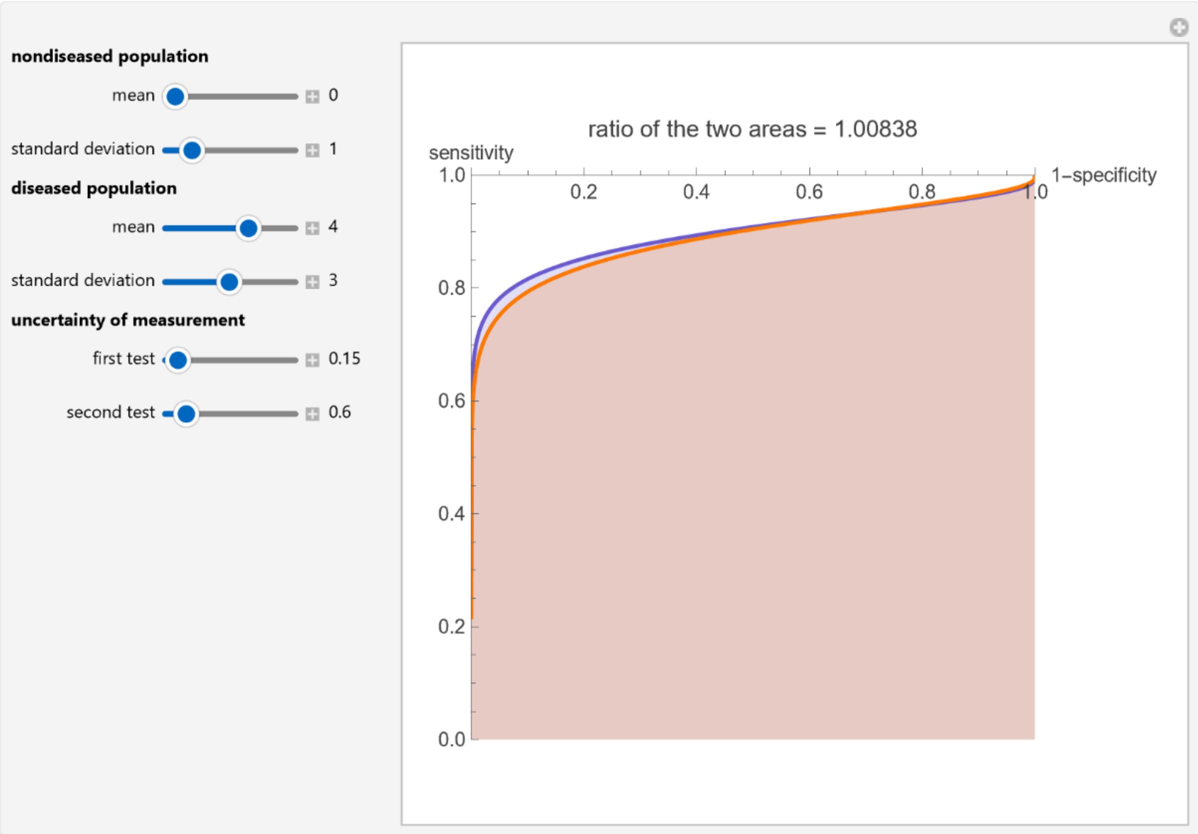

Figure 4: *Receiver operating characteristic (ROC) plots of two diagnostic tests (first test: blue plot, second test: orange plot), and the ratio of the respective areas under the curves, with the settings shown at the left.*

Details

ROC curves are used in the evaluation of the clinical accuracy of a diagnostic test applied to a diseased and a nondiseased population. They display sensitivity versus 1-specificity. Sensitivity is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test. Therefore, the ROC plots display the true positive fraction versus the false positive fraction. Furthermore, the area under a ROC curve is used as an index of the diagnostic accuracy of the respective test.

This Demonstration could be useful in exploring ROC curves and evaluating the maximum medically permissible uncertainty of measurement of a diagnostic test. For example, in the second figure the population data describe a bimodal distribution of serum glucose measurements in a nondiabetic and a diabetic population [1].

Reference

[1] T. O. Lim, R. Bakri, Z. Morad, and M. A. Hamid. Bimodality in Blood Glucose Distribution: Is It Universal? *Diabetes Care,* **25**(12), 2002 pp. 2212–2217.

Demonstration

Version

1.1.2

DOI

10.5281/zenodo.20535253

Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/ReceiverOperatingCharacteristicCurvesAndUncertaintyOfMeasure.nb

First published on June 12, 2007, as part of the Wolfram Demonstrations Project.

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS
Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.
System memory (RAM): 4 GB+ recommended.

Citation

Hatjimihail AT. *Receiver Operating Characteristic Curves and Uncertainty of Measurement*. Ver 1.1.2. Hellenic Complex Systems Laboratory; 2026. https://doi.org/10.5281/zenodo.20535253

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Technical Report

DOI

10.5281/zenodo.20535267

Citation

Hatjimihail AT. *Receiver Operating Characteristic Curves and Uncertainty of Measurement.* Technical Report V. Hellenic Complex Systems Laboratory; 2007. https://doi.org/10.5281/zenodo.20535267

First Published: June 12, 2007

Revised: May 23, 2026